

States Of Matter Gases And Liquids

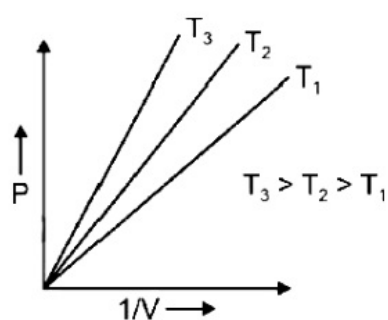
Question1

Which amongst the following options are correct graphical representation of Boyle's law?

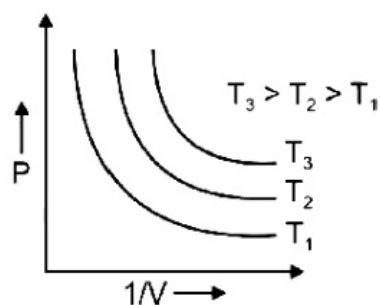
[NEET 2023]

Options:

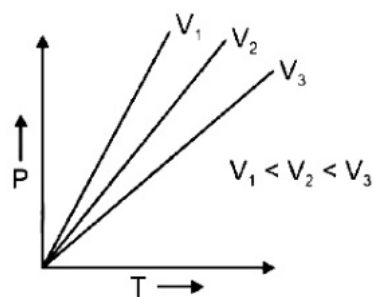
A.



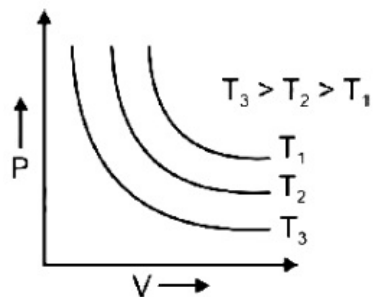
B.



C.



D.



Answer: A

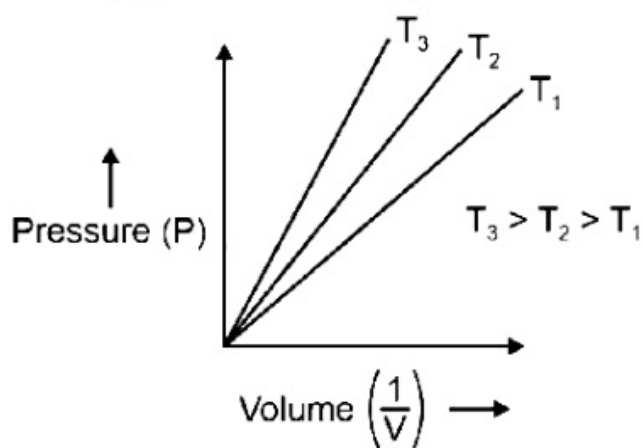
Solution:

According to Boyle's law,

$$PV = nRT$$

$$P = nRT \left(\frac{1}{V} \right)$$

P versus $\left(\frac{1}{V} \right)$ gives straight line graph with slope nRT .



Question2

Intermolecular forces are forces of attraction and repulsion between interacting particles that will include :

- A. dipole - dipole forces**
- B. dipole - induced dipole forces**
- C. hydrogen bonding**
- D. covalent bonding**
- E. dispersion forces**

Choose the most appropriate answer from the options given below :

[NEET 2023]

Options:

A.

A, B, C, D are correct

B.

A, B, C, E are correct

C.

A, C, D, E are correct

D.

B, C, D, E are correct

Answer: B

Solution:

Intermolecular forces are the forces of attraction and repulsion between interacting molecules. This term does not include covalent bonds as covalent bond holds atoms of a molecule together.

Hence, dipole - dipole forces, dipole - induced dipole forces, hydrogen bonding and dispersion forces are intermolecular forces.

Question3

The correct van der Waals equation for 1 mole of a real gas is :

[NEET 2023 mpr]

Options:

A.

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

B.

$$\left(p + \frac{V^2}{a}\right)(V - b) = RT$$

C.

$$\left(p + \frac{an^2}{V^2}\right)(V^2 - nb) = RT$$

D.

$$\left(p + \frac{an^2}{V}\right)(V - nb) = nRT$$

Answer: A

Solution:

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$

Question4

The correct option in which the density of argon (Atomic mass = 40) is highest :

[NEET 2023 mpr]

Options:

A.

STP

B.

0°C, 2atm

C.

0°C, 4atm

D.

273°C, 4atm

Answer: C

Solution:

$$\rho = PM/RT$$

For maximum density, pressure should be maximum and temperature should be minimum.

Question5

Which one is not correct mathematical equation for Dalton's Law of partial pressure? Here p = total pressure of gaseous mixture
[NEET-2022]

Options:

A. $p = p_1 + p_2 + p_3$

B. $p = n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$

C. $p_i = \chi_i p$, where p_i = partial pressure of i^{th} gas
 χ_i = mole fraction of i^{th} gas in gaseous mixture

D. $p_i = \chi_i p_i^0$, χ_i = mole fraction of i^{th} gas in gaseous mixture
 p_i^0 = pressure of i^{th} gas in pure state

Answer: D

Solution:

-Dalton's law of partial pressure states that the total pressure by the mixture of non-reactive gases is equal to the sum of the partial pressures of individual gases.

$$- p_{\text{Total}} = p_1 + p_2 + p_3$$

- Also, $p_i = \chi_i p$; where p_i and χ_i are the partial pressure and mole fraction of i^{th} gas respectively and p is the total pressure.

$$- p_{\text{Total}} = p_1 + p_2 + p_3$$

$$= n_1 \frac{RT}{V} + n_2 \frac{RT}{V} + n_3 \frac{RT}{V}$$

$$= (n_1 + n_2 + n_3) \frac{RT}{V}$$

Question6

A 10.0L flask contains 64g of oxygen at 27°C. (Assume O₂ gas is behaving ideally). The pressure inside the flask in bar is (Given R = 0.0831L bar K⁻¹mol⁻¹)
[NEET-2022]

Options:

A. 2.5

B. 498.6

C. 49.8

D. 4.9

Answer: D

Solution:

We know for ideal gas

$$PV = nRT$$

$$P = n \frac{RT}{V}$$

$$P = \frac{64}{32} \times \frac{0.0831 \times 300}{10}$$

$$P = 4.9 \text{ bar}$$

Pressure of O₂ gas inside the flask = 4.9 bar

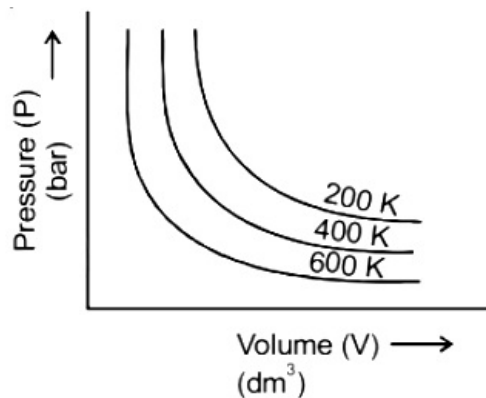
Question7

Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures :

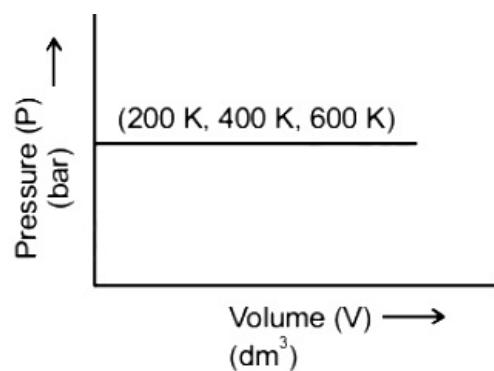
[NEET 2021]

Options:

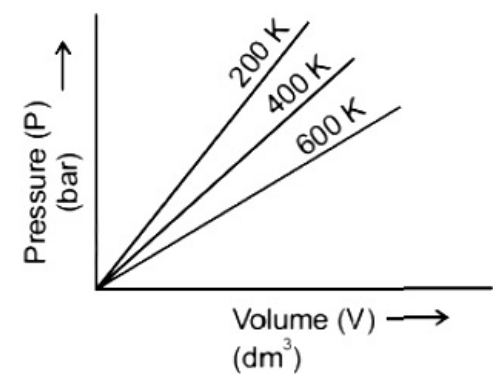
A.



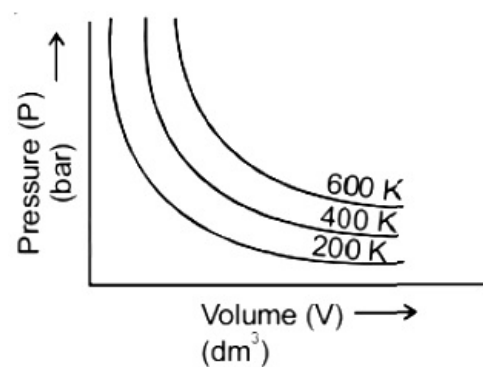
B.



C.



D.



Answer: D

Solution:

According to Boyle's law

$$P \propto \frac{1}{V} \Rightarrow P = \frac{k}{V} \Rightarrow PV = k$$

where k is proportionality constant and equal to nRT.

∴ Graph between P vs. V should be rectangular hyperbola and product of PV increases with increase in temperature.

Question8

Choose the correct option for the total pressure (in atm.) in a mixture of 4gO₂ and 2gH₂ confined in a total volume of one litre at 0°C is :

[Given R = 0.082L atm mol⁻¹K⁻¹, T = 273 K]

[NEET 2021]

Options:

A. 2.518

B. 2.602

C. 25.18

D. 26.02

Answer: C

Solution:

$$n_{O_2} = \frac{4}{32} = \frac{1}{8}$$

$$n_{H_2} = \frac{2}{2} = 1$$

$$n_t = \frac{1}{8} + 1 = \frac{9}{8}$$

$$P_t V = n_t RT$$

$$P_t = \frac{\frac{9}{8} \times 0.082 \times 273}{1} = 25.18 \text{ atm}$$

Question9

A mixture of N₂ and Ar gases in a cylinder contains 7g of N₂ and 8g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N₂ is:

**[Use atomic masses (in gmol⁻¹) : N = 14, Ar = 40]
(2020)**

Options:

- A. 12 bar
- B. 15 bar
- C. 18 bar
- D. 9 bar

Answer: B

Solution:

$$(b) \ n_{N_2} = \frac{7}{28} = \frac{1}{4} = 0.25$$

$$n_{Ar} = \frac{8}{40} = \frac{1}{5} = 0.20$$

Now, applying Dalton's law of partial pressure,

$$p_{N_2} = \text{mole fraction of } N_2 \cdot P_{\text{Total}}$$

$$\frac{0.25}{0.45} \times 27 = \frac{5}{9} \times 27 = 15 \text{ bar}$$

Question10

A gas at 350 K and 15 bar has molar volume 20 percent smaller than that for an ideal gas under the same conditions. The correct option about the gas and its compressibility factor (Z) is (NEET 2019)

Options:

- A. $Z < 1$ and repulsive forces are dominant
- B. $Z > 1$ and attractive forces are dominant
- C. $Z > 1$ and repulsive forces are dominant
- D. $Z < 1$ and attractive forces are dominant.

Answer: D

Solution:

Solution:

$$V_{\text{ideal}} = V, V_{\text{real}} = V - 0.2V = 0.8V$$

$$Z = \frac{V_{\text{real}}}{V_{\text{ideal}}} = 0.8$$

if value of $Z < 1$ then attractive forces are dominant.

Question11

The volume occupied by 1.8g of water vapour at 374°C and 1 bar

**pressure will be [Use $R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$]
(Odisha NEET 2019)**

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Options:

- A. 96.66L
- B. 55.87L
- C. 3.10L
- D. 5.37L

Answer: D

Solution:

Solution:

$$m = 1.8\text{g}$$

$$n = \frac{m}{M} = \frac{1.8}{18} = 0.1\text{mol}$$

$$T = 374^\circ\text{C} = 647\text{K}, P = 1\text{bar}$$

$$R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$$

$$V = \frac{nRT}{P} = \frac{0.1 \times 0.083 \times 647}{1} = 5.37\text{L}$$

Question12

**Given van der Waals' constant for N H_3 , H_2 , O_2 and CO_2 are respectively 4.17, 0.244, 1.36 and 3.59, which one of the following gases is most easily liquefied?
(NEET 2018)**

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Options:

- A. N H_3
- B. H_2
- C. O_2
- D. CO_2

Answer: A

Solution:

Solution:

van der Waals' constant 'a' signifies the intermolecular forces of attraction between the particle of gas. So, higher the value of 'a', easier will be the liquefaction of gas.

Question13

Equal moles of hydrogen and oxygen gases are placed in a container with a pin-hole through which both can escape. What fraction of the oxygen escapes in the time required for one-half of the hydrogen to escape?

(NEET-I 2016)

Options:

A. $\frac{3}{8}$

B. $\frac{1}{2}$

C. $\frac{1}{8}$

D. $\frac{1}{4}$

Answer: C

Solution:

Solution:

Let the number of moles of each gas = x

Fraction of hydrogen escaped = $\frac{1}{2}x$

$$\frac{r_{O_2}}{r_{H_2}} = \sqrt{\frac{M_{H_2}}{M_{O_2}}} \Rightarrow \frac{n_{O_2}/t}{\frac{x}{2}/t} = \sqrt{\frac{2}{32}} = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

$$\Rightarrow \frac{n_{O_2}/t}{\frac{x}{2}/t} = \frac{1}{4} \Rightarrow n_{O_2} = \frac{1}{8}x$$

Hence, fraction of oxygen escaped = $\frac{1}{8}$

Question14

A gas such as carbon monoxide would be most likely to obey the ideal gas law at
(2015)

Options:

A. low temperatures and high pressures

B. high temperatures and high pressures

C. low temperatures and low pressures

D. high temperatures and low pressures

Answer: D

Solution:

Real gases show ideal gas behaviour at high temperatures and low pressures

Question15

Maximum deviation from ideal gas is expected from (2013 NEET)

Options:

A. $\text{CH}_{4(g)}$

B. $\text{NH}_{3(g)}$

C. $\text{H}_{2(g)}$

D. $\text{N}_{2(g)}$

Answer: B

Solution:

Solution:

NH_3 is a polar molecule, thus more attractive forces between its molecule.

Question16

**What is the density of N_2 gas at 227°C and 5.00 atm pressure?
($R = 0.082\text{L atm K}^{-1}\text{mol}^{-1}$)
(Karnataka NEET 2013)**

Options:

A. 1.40g/mL

B. 2.81g/mL

C. 3.41g/mL

D. 0.29g/mL

Answer: C

Solution:

$$PV = nRT$$

$$PV = \frac{W}{M}RT \left[n = \frac{\text{Weight of the gas taken (W)}}{\text{Mol. mass of gas (M)}} \right]$$

$$P = \frac{W}{M} \times \frac{RT}{V}$$

$$P = \frac{dRT}{M} \left[\text{Density} = \frac{\text{Mass}}{\text{Volume}} \right]$$

$$d = \frac{PM}{RT} = \frac{5 \times 28}{0.0821 \times 500} = 3.41 \text{ g/mL}$$

Question 17

For real gases van der Waals equation is written as

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT$$

where a and b are van der Waals constant

Two sets of gases are

(1) O_2 , CO_2 , H_2 and He (2) CH_4 , O_2 and H_2

The gases given in set-I in increasing order of b and gases given in set-II in decreasing order of a, are arranged below.

Select the correct order from the following

(2012 Mains)

Options:

A.

(1) $\text{He} < \text{H}_2 < \text{CO}_2 < \text{O}_2$ (2) $\text{CH}_4 > \text{H}_2 > \text{O}_2$

B.

(1) $\text{O}_2 < \text{He} < \text{H}_2 < \text{CO}_2$ (2) $\text{H}_2 > \text{O}_2 > \text{CH}_4$

C.

(1) $\text{H}_2 < \text{He} < \text{O}_2 < \text{CO}_2$ (2) $\text{CH}_4 > \text{O}_2 > \text{H}_2$

D.

(1) $\text{H}_2 < \text{O}_2 < \text{He} < \text{CO}_2$ (2) $\text{O}_2 > \text{CH}_4 > \text{H}_2$

Answer: C

Solution:

Van der Waal gas constant 'a' represent intermolecular force of attraction of gaseous molecules and Van der Waal gas constant 'b' represent effective size of molecules. Therefore order should be

(1) $H_2 < He < O_2 < CO_2$

(2) $CH_4 > O_2 > H_2$

Question18

A certain gas takes three times as long to effuse out as helium. Its molecular mass will be (2012 Mains)

Options:

A. 27 u

B. 36 u

C. 64 u

D. 9 u

Answer: B

Solution:

According to Graham's law of diffusion

$$r \propto \frac{1}{\sqrt{d}} \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

$$\text{Rate of diffusion} = \frac{\text{Volume of gas diffused (V)}}{\text{Time taken (t)}}$$

$$\therefore \frac{V_1/t_1}{V_2/t_2} = \sqrt{\frac{M_2}{M_1}}$$

If same volume of two gases diffuse, then $V_1 = V_2$

$$\therefore \frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}}$$

Here $t_2 = 3t_1$, $M_1 = 4u$, $M_2 = ?$

$$\therefore \frac{3t_1}{t_1} = \sqrt{\frac{M_2}{4}} \Rightarrow 3 = \sqrt{\frac{M_2}{4}}$$

$$\Rightarrow 9 = \frac{M_2}{4} \Rightarrow M_2 = 36u$$

Question19

Equal volumes of two monatomic gases, A and B at same temperature and pressure are mixed. The ratio of specific heats (C_p/C_v) of the mixture will be (2012)

Options:

- A. 0.83
- B. 1.50
- C. 3.3
- D. 1.67

Answer: D**Solution:****Solution:**

C_p for monoatomic gas mixture of same volume = $\frac{5}{2}R$

$$C_v = \frac{3}{2}R$$

$$\therefore \frac{C_p}{C_v} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = \frac{5}{3} = 1.67$$

Question20

50 mL of each gas A and of gas B takes 150 and 200 seconds respectively for effusing through a pin hole under the similar conditions. If molecular mass of gas B is 36, the molecular mass of gas A will be (2012)

Options:

- A. 96
- B. 128
- C. 32
- D. 64
- E. None

Answer: E**Solution:**

(None): According to Graham's law of diffusion,

$$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}}$$

$$r_A = \frac{V_A}{T_A}, r_B = \frac{V_B}{T_B}$$

$$\frac{V_A/T_A}{V_B/T_B} = \sqrt{\frac{M_B}{M_A}}$$

$$V_A = V_B, T_A = 150\text{sec}, T_B = 200\text{sec}$$

$$M_B = 36, M_A = ?$$

$$\frac{T_B}{T_A} = \sqrt{\frac{M_B}{M_A}} \Rightarrow \frac{200}{150} = \sqrt{\frac{36}{M_A}}$$

$$\frac{4}{3} = \sqrt{\frac{36}{M_A}} \text{ or } \frac{4 \times 4}{3 \times 3} = \frac{36}{M_A}$$

$$\text{or } M_A = \frac{36}{4 \times 4} \times 3 \times 3 = 20.25$$

No answer is correct.

Question21

A bubble of air is underwater at temperature 15°C and the pressure 1.5 bar. If the bubble rises to the surface where the temperature is 25°C and the pressure is 1.0 bar, what will happen to the volume of the bubble?

(2011 Mains)

Options:

- A. Volume will become greater by a factor of 1.6
- B. Volume will become greater by a factor of 1.1
- C. Volume will become smaller by a factor of 0.70
- D. Volume will become greater by a factor of 2.5

Answer: A

Solution:

We know that ideal gas equation, $V \propto \frac{T}{P}$

Given $T_1 = 15 + 273 = 288$, $P_1 = 1.5\text{bar}$

$T_2 = 25 + 273 = 298$, $P_2 = 1\text{bar}$

$$V_1 \propto \frac{288}{1.5} \text{ i.e. } V_1 \propto 192$$

$$V_2 \propto \frac{298}{1}$$

$$\frac{V_2}{V_1} = \frac{298}{192} = 1.55 \approx 1.6$$

Question22

A gaseous mixture was prepared by taking equal mole of CO and N₂. If the total pressure of the mixture was found 1 atmosphere, the partial

pressure of the nitrogen (N_2) in the mixture is (2011)

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Options:

- A. 0.5 atm
- B. 0.8 atm
- C. 0.9 atm
- D. 1 atm

Answer: A

Solution:

$$P_{\text{CO}} + P_{\text{N}_2} = 1\text{atm}$$

$$2P_{\text{N}_2} = 1 \quad [\because n_{\text{CO}} = n_{\text{N}_2}]$$

$$P_{\text{N}_2} = \frac{1}{2} = 0.5\text{atm}$$

Question23

By what factor does the average velocity of a gaseous molecule increase when the temperature (in Kelvin) is doubled? (2011)

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Options:

- A. 2.0
- B. 2.8
- C. 4.0
- D. 1.4

Answer: D

Solution:

$$\text{Average velocity} = \sqrt{\frac{8RT}{\pi M}}$$

Question24

Two gases A and B having the same volume diffuse through a porous partition in 20 and 10 seconds respectively. The molecular mass of A is 49 u. Molecular mass of B will be (2011)

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Options:

- A. 50.00 u
- B. 12.25 u
- C. 6.50 u
- D. 25.00 u

Answer: B

Solution:

Solution:

We know that $\frac{r_A}{r_B} = \frac{v/t_A}{v/t_B} = \sqrt{\frac{M_B}{M_A}}$

$$\frac{t_B}{t_A} = \sqrt{\frac{M_B}{M_A}} \Rightarrow \frac{10}{20} = \sqrt{\frac{M_B}{49}}$$

$$\Rightarrow \left(\frac{10}{20}\right)^2 = \frac{M_B}{49} \Rightarrow \frac{100}{400} = \frac{M_B}{49}$$

$$\Rightarrow M_B = \frac{49 \times 100}{400} = 12.25u$$

Question25

The pressure exerted by 6.0 g of methane gas in a 0.03m³ vessels at 129°C is (Atomic masses:C = 12.01,H = 1.01 and R = 8.314J K⁻¹mol⁻¹) (2010 Mains)

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Options:

- A. 215216 Pa
- B. 13409 Pa
- C. 41648 Pa
- D. 31684 Pa

Answer: C

Solution:

Solution:

Given, mass of CH₄, w = 6g

Volume of CH_4 , $V = 0.03\text{m}^3$

$T = 129^\circ = 129 + 273 = 402\text{K}$

$R = 8.314\text{J K}^{-1}\text{mol}^{-1}$

Molecular mass of

CH_4 , $M = 12.01 + 4 \times 1.01 = 16.05$

$PV = nRT = \frac{w}{M}RT$

$\therefore P = \frac{w}{M} \frac{RT}{V} = \frac{6}{16.05} \times \frac{8.314 \times 402}{0.03}$

$= 41647.7\text{Pa} \approx 41648\text{Pa}$

Question26

The energy absorbed by each molecule (A_2) of a substance is $4.4 \times 10^{-19}\text{J}$ and bond energy per molecule is $4.0 \times 10^{-19}\text{J}$ The kinetic energy of the molecule per atom will be (2009)

Options:

A. $2.2 \times 10^{-19}\text{J}$

B. $2.0 \times 10^{-19}\text{J}$

C. $4.0 \times 10^{-20}\text{J}$

D. $2.0 \times 10^{-20}\text{J}$

Answer: D

Solution:

Solution:

Energy absorbed by each molecule = $4.4 \times 10^{-19}\text{J}$

Energy required to break the bond = $4.0 \times 10^{-19}\text{J}$

Remaining energy gets converted to kinetic energy

$= (4.4 \times 10^{-19} - 4.0 \times 10^{-19})\text{J}$

$= 0.4 \times 10^{-19}\text{J}$ per molecule

$\therefore$ Kinetic energy per atom = $0.2 \times 10^{-19}\text{J} = 2 \times 10^{-20}\text{J}$

Question27

If a gas expands at constant temperature, it indicates that (2008)

Options:

A. kinetic energy of molecules remains the same

B. number of the molecules of gas increases

C. kinetic energy of molecules decreases

D. pressure of the gas increases

Answer: A

Solution:

The average translational K.E. of one molecule of an ideal gas will be given by

$$E_t = \frac{K \cdot E}{N_A} = \frac{3/2 RT}{N_A} = \frac{3}{2} K T$$

When $\frac{R}{N_A}$ = Boltzmann constant i.e. $E_t \propto T$

So, at constant temperature K.E. of molecules remains same,

Question28

Volume occupied by one molecule of water (density = 1gcm^{-3}) is (2008)

Options:

A. $3.0 \times 10^{-23}\text{cm}^3$

B. $5.5 \times 10^{-23}\text{cm}^3$

C. $9.0 \times 10^{-23}\text{cm}^3$

D. $6.023 \times 10^{-23}\text{cm}^3$

Answer: A

Solution:

Solution:

Wt. of 6.023×10^{23} molecule of water = 18g

$\therefore$ Volume occupied by 6.023×10^{23} molecule of water (density = 1gcm^{-3}) will be

$$= \frac{18\text{g}}{1\text{gcm}^{-3}} = 18\text{cm}^3 \text{ or ml}$$

$\therefore$ Volume occupied by one molecule of water

$$= \frac{18}{6.023 \times 10^{23}} = 2.988 \times 10^{-23} = 3.0 \times 10^{-23}\text{cm}^3$$

Question29

van der Waals' real gas, acts as an ideal gas, at which conditions? (2002)

Options:

- A. High temperature, low pressure
- B. Low temperature, high pressure
- C. High temperature, high pressure
- D. Low temperature, low pressure (2002)

Answer: A

Solution:

At low pressure and high temperature van der Waals real gas acts as ideal gas and observed to obey $PV = nRT$ relation. At very low pressure when the gas-volume is quite large the space occupied by the molecules themselves becomes negligible comparatively and because the molecules are then far apart, the force of mutual attraction becomes too feeble, the real gas would satisfy the postulates of kinetic theory. As temperature is raised, the volume of the gas increases and we can consider

$\left(P + \frac{n^2 a}{V^2} \right)$ term as P and at low pressure $(V - nb)$ term as V

$\left(P + \frac{n^2 a}{V^2} \right) (V - nb) = nRT$ (van der Waals' equation)

The gas equation becomes $PV = nRT$
This is ideal gas equation.

Question30

Average molar kinetic energy of CO and N_2 at same temperature is (2000)

Options:

- A. $KE_1 = KE_2$
- B. $KE_1 > KE_2$
- C. $KE_1 < KE_2$
- D. can't say anything. Both volumes are not given.

Answer: A

Solution:

Solution:

$KE = \frac{3}{2}RT$ (for one mole of a gas)

As temperatures are same and KE is independent of molecular mass, so $KE_1 = KE_2$.

Question31

At 25°C and 730mm pressure, 380mL of dry oxygen was collected. If the temperature is constant, what volume will the oxygen occupy at 760mm pressure?
(1999)

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Options:

- A. 569mL
- B. 365mL
- C. 265mL
- D. 621mL

Answer: B

Solution:

$$V_1 = 380\text{mL}, P_1 = 730\text{mm}, V_2 = ?$$

$$P_2 = 760\text{mm}$$

$$\text{From Boyle's law, } P_1 V_1 = P_2 V_2$$

$$\Rightarrow V_2 = \frac{380 \times 730}{760} = 365\text{mL}$$

Question32

Which of the following statements is wrong for gases?
(1999)

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Options:

- A. Confined gas exerts uniform pressure on the walls of its container in all directions.
- B. Volume of the gas is equal to volume of container confining the gas.
- C. Gases do not have a definite shape and volume.
- D. Mass of a gas cannot be determined by weighing a container in which it is enclosed.

Answer: D

Solution:

Solution:

Mass of the gas = mass of the cylinder including gas - mass of empty cylinder.

So mass of a gas can be determined by weighing the container in which it is enclosed.

Thus, the statement (d) is wrong for gases.

Question33

The average kinetic energy of an ideal gas, per molecule in S.I. units, at 25°C will be (1996)

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Options:

A. $6.17 \times 10^{-20}\text{J}$

B. $7.16 \times 10^{-20}\text{J}$

C. $61.7 \times 10^{-20}\text{J}$

D. $6.17 \times 10^{-21}\text{J}$

Answer: D

Solution:

Temperature (T) = 25°C = 298K

Therefore K. E. per molecule

$$= \frac{3RT}{2N_A} = \frac{3 \times 8.314 \times 298}{2 \times (6.02 \times 10^{23})} = 6.17 \times 10^{-21}\text{J}$$

Question34

At what temperature, the rate of effusion of N₂ would be 1.625 times than the rate of SO₂ at 500°C? (1996)

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Options:

A. 373°C

B. 620°C

C. 110°C

D. 173°C

Answer: B

Solution:

Solution:

$$R_1 = 1.625R_2 \text{ and } T_2 = 500^\circ\text{C} = 773\text{K}$$

We know that $\frac{R_1}{R_2} = \sqrt{\frac{M_1}{M_2} \times \frac{T_1}{T_2}}$

$$\text{or } 1.625 = \sqrt{\frac{64}{28} \times \frac{T_1}{773}}$$

or

$$T_1 = \frac{(1.625)^2 \times 28 \times 773}{64} = 893\text{K} = 620^\circ\text{C}$$

Question35

**Which of the following mixture of gases does not obey Dalton's law of partial pressure?
(1996)**

Options:

A. Cl_2 and SO_2

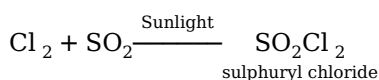
B. CO_2 and He

C. O_2 and CO_2

D. N_2 and O_2

Answer: A

Solution:



Dalton's law of partial pressure is applicable only in those cases where gases are non-reacting. As Cl_2 and SO_2 reacts to form SO_2Cl_2 so this law is not obeyed in given case.

Question36

**An ideal gas, obeying kinetic theory of gases can not be liquefied, because
(1995)**

Options:

A. it solidifies before becoming a liquid

B. forces acting between its molecules are negligible

C. its critical temperature is above 0°C

D. its molecules are relatively small in size.

Answer: B

Solution:

A gas can only be liquefied, if some forces of attraction are acting in its molecules. since, an ideal gas is devoid of force of attraction in its molecules, therefore it can not be liquefied.

Question37

50 mL of hydrogen diffuses out through a small hole of a vessel, in 20 minutes. The time taken by 40mL of oxygen to diffuse out is (1994)

Options:

- A. 32 minutes
- B. 64 minutes
- C. 8 minutes
- D. 12 minutes

Answer: B

Solution:

Volume of hydrogen = 50mL;

Time for diffusion (t) = 20 min and volume of oxygen = 40mL

Rate of diffusion of hydrogen (r_1) = $\frac{50}{20} = 2.5\text{mL/min}$

Rate of diffusion of oxygen (r_2) = $\frac{40}{t}\text{mL/min}$

since the molecular mass of hydrogen (M_1) = 2 and that of oxygen (M_2) = 32, therefore

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} \Rightarrow \frac{2.5}{40/t} = \sqrt{\frac{32}{2}}$$

$$\Rightarrow \frac{t}{16} = 4 \Rightarrow t = 64 \text{ minutes}$$

Question38

The temperature of a gas is raised from 27°C to 927°C. The root mean square speed of the gas (1994)

Options:

A. remains same

B. gets $\sqrt{\frac{927}{27}}$ times

C. gets halved

D. gets doubled.

Answer: D

Solution:

$T_1 = 27^\circ\text{C} = 300\text{K}$ and

$T_2 = 927^\circ\text{C} = 1200\text{K}$

We know that root mean square speed (v) $\propto \sqrt{T}$.

Therefore root mean square speed of the gas, when

its temperature is raised $= \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{1200}{300}} = 2$

Question39

Under what conditions will a pure sample of an ideal gas not only exhibit a pressure of 1 atm but also a concentration of 1 mole litre⁻¹ ? (R = 0.082 litre atm mol⁻¹deg⁻¹) (1993)

Options:

A. At STP

B. When $V = 22.4$ litres

C. When $T = 12\text{K}$

D. Impossible under any conditions

Answer: C

Solution:

$PV = nRT$ or $P = \frac{n}{V}RT = CRT$

Hence,

$1 = 1 \times 0.082 \times T \Rightarrow T = \frac{1}{0.082} = 12\text{K}$

Question40

Internal energy and pressure of a gas per unit volume are related as (1993)

Options:

A. $P = \frac{2}{3}E$

B. $P = \frac{3}{2}E$

C. $P = \frac{1}{2}E$

D. $P = 2E$

Answer: A

Solution:

$$\begin{aligned} PV &= \frac{1}{3}mnu^2 = \frac{1}{3}Mu^2 \\ &= \frac{2}{3} \cdot \frac{1}{2}Mu^2 = \frac{2}{3}E \quad \left\{ \because \frac{1}{2}Mu^2 = E \right\} \\ \text{or } P &= \frac{2}{3}E \text{ per unit volume.} \end{aligned}$$

Question41

The ratio among most probable velocity, mean velocity and root mean square velocity is given by (1993)

Options:

A. 1 : 2 : 3

B. $1 : \sqrt{2} : \sqrt{3}$

C. $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$

D. $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$

Answer: D

Solution:

Most probable velocity, $(u_{mp}) = \sqrt{\frac{2RT}{M}}$

Mean velocity, $(\bar{v}) = \sqrt{\frac{8RT}{\pi M}}$

$$\begin{aligned} \text{Root mean square velocity, } (u_{\text{r.m.s}}) &= \sqrt{\frac{3RT}{M}} \\ \therefore u_{\text{mp}} : \bar{v} : u_{\text{r.m.s}} &= \sqrt{\frac{2RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{3RT}{M}} \\ &= \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3} \end{aligned}$$

Question42

At STP, 0.50mol H₂ gas and 1.0mol He gas (1993)

Options:

- A. have equal average kinetic energies
- B. have equal molecular speeds
- C. occupy equal volumes
- D. have equal effusion rates.

Answer: A

Solution:

Solution:

Because average kinetic energy depends only on temperature $K E = \frac{3}{2}nkT$

Question43

When is deviation more in the behaviour of a gas from the ideal gas equation $PV = nRT$? (1993)

Options:

- A. At high temperature and low pressure
- B. At low temperature and high pressure
- C. At high temperature and high pressure
- D. At low temperature and low pressure

Answer: B

Solution:

At low temperature and high pressure, there is a deviation from the ideal behaviour in gases.

Question44

A closed flask contains water in all its three states solid, liquid and vapour at 0°C . In this situation, the average kinetic energy of water molecules will be (1992)

Options:

- A. the greatest in all the three states
- B. the greatest in vapour state
- C. the greatest in the liquid state
- D. the greatest in the solid state.

Answer: B

Solution:

Solution:

Velocity and hence average K . E . of water molecules is maximum in the gaseous state.

Question45

Which is not true in case of an ideal gas? (1992)

Options:

- A. It cannot be converted into a liquid.
- B. There is no interaction between the molecules.
- C. All molecules of the gas move with same speed.
- D. At a given temperature, PV is proportional to the amount of the gas.

Answer: C

Solution:

Solution:

Molecules in an ideal gas move with different speeds. Due to collision between the particle their speed changes.

Question46

The correct value of the gas constant R ' is close to (1992)

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Options:

- A. 0.082 litre-atmosphere K
- B. 0.082 litre-atmosphere $\text{K}^{-1}\text{mol}^{-1}$
- C. 0.082 litre-atmosphere $^{-1}\text{K mol}^{-1}$
- D. 0.082 litre $^{-1}$ atmosphere $^{-1}\text{K mol}$.

Answer: B

Question47

An ideal gas can't be liquefied because (1992)

Options:

- A. its critical temperature is always above 0°C
- B. its molecules are relatively smaller in size
- C. it solidifies before becoming a liquid
- D. forces operative between its molecules are negligible.

Answer: D

Solution:

Solution:

In an ideal gas, the intermolecular forces of attraction are negligible and hence it cannot be liquefied.

Question48

Select one correct statement. In the gas equation, $PV = nRT$ (1992)

Options:

- A. n is the number of molecules of a gas
- B. V denotes volume of one mole of the gas
- C. n moles of the gas have a volume V
- D. P is the pressure of the gas when only one mole of gas is present.

Answer: C

Solution:**Solution:**

In the van der Waals' equation
 $PV = nRT$ n
moles of the gas have volume V

Question49

A gas is said to behave like an ideal gas when the relation $PV / T = \text{constant}$. When do you expect a real gas to behave like an ideal gas? (1991)

Options:

- A. When the temperature is low.
- B. When both the temperature and pressure are low.
- C. When both the temperature and pressure are high.
- D. When the temperature is high and pressure is low.

Answer: D

Solution:**Solution:**

At high temperature and low pressure the effect of a/V^2 and b is negligible.
As we know, $PV = nRT$ (Ideal gas equation)
 $PV = nRT$ or $\frac{PV}{nRT} = 1$
 $\therefore Z = 1$ [Z is compressibility factor]
Hence gas shows ideal behaviour.

Question50

**In a closed flask of 5 litres, 1.0g of H₂ is heated from 300 to 600K .
Which statement is not correct?
(1991)**

Options:

- A. Pressure of the gas increases
- B. The rate of collision increases
- C. The number of moles of gas increases
- D. The energy of gaseous molecules increases

Answer: C

Solution:

Here volume is constant and mass of H₂ is fixed so the no. of moles of the gas do not change. As temperature increases the pressure also increases, therefore the rate of collision among the gas molecules and their energy also increases. Mass is fixed. It won't be effected by increase in temperature in a closed system. Therefore no. of moles of gas is also fixed. Hence option C is an incorrect statement.

Question51

**At constant temperature, in a given mass of an ideal gas
(1991)**

Options:

- A. the ratio of pressure and volume always remains constant
- B. volume always remains constant
- C. pressure always remains constant
- D. the product of pressure and volume always remains constant.

Answer: D

Solution:

Solution:

According to Boyle's law at constant temperature, $P \propto \frac{1}{V}$ or $PV = \text{constant}$

Question52

The root mean square velocity at STP for the gases H₂, N₂, O₂ and HBr are in the order (1991)

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Options:

A. H₂ < N₂ < O₂ < HBr

B. HBr < O₂ < N₂ < H₂

C. H₂ < N₂ = O₂ < HBr

D. HBr < O₂ < H₂ < N₂

Answer: B

Solution:

Solution:

We know, $PV = \frac{1}{3}mnu^2 = \frac{1}{3}Mu^2$

or $u = \sqrt{3PV/M}$

At STP, $u \propto \sqrt{\frac{1}{M}}$

and molecular masses of H₂, N₂, O₂ and HBr are 2, 28, 32 and 81

Question 53

Root mean square velocity of a gas molecule is proportional to (1990)

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Options:

A. $m^{1/2}$

B. m^0

C. $m^{-1/2}$

D. m

Answer: C

Solution:

Solution:

$PV = \frac{1}{3}mNu^2$

here u = root mean square velocity.

Now $u^2 = \frac{3PV}{mN}$ or $u \propto \frac{1}{\sqrt{m}}$

Question54

Absolute zero is defined as the temperature (1990)

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Options:

- A. at which all molecular motion ceases
- B. at which liquid helium boils
- C. at which ether boils
- D. all of the above.

Answer: A

Solution:

Solution:

Temperature at which all molecular motion ceases is called absolute zero.

Question55

In van der Waals equation of state for a non-ideal gas, the term that accounts for intermolecular forces is (1990)

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Options:

- A. $(V - b)$
- B. $(RT)^{-1}$
- C. $\left(P + \frac{a}{V^2}\right)$
- D. RT

Answer: C

Solution:

Solution:

van der Waals' equation for 1 mole is $\left(P + \frac{a}{V^2}\right)(V - b) = RT$

Here, $\left(P + \frac{a}{V^2}\right)$ represents the intermolecular forces and $(V - b)$ is the corrected volume.

Question56

If P, V, M, T and R are pressure, volume, molar mass, temperature and gas constant respectively, then for an ideal gas, the density is given by (1989)

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Options:

A. $\frac{RT}{PM}$

B. $\frac{P}{RT}$

C. $\frac{M}{V}$

D. $\frac{PM}{RT}$

Answer: D

Solution:

Ideal gas equation is

$$PV = nRT = \frac{m}{M}RT$$

$$\text{or } PM = \frac{m}{V}RT = d RT \quad [\text{here } d = \text{density}]$$

$$\Rightarrow d = \frac{PM}{RT}$$

Question57

Pressure remaining the same, the volume of a given mass of an ideal gas increases for every degree centigrade rise in temperature by definite fraction of its volume at (1989)

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Options:

A. 0°C

B. its critical temperature

C. absolute zero

D. its Boyle temperature.

Answer: A

Solution:

According to Charles' law which states that "The volume of the given mass of a gas increases or decreases by $\frac{1}{273}$ of its volume at 0°C for each degree rise or fall of temperature at constant pressure."

$$V_t = V_0 \left(1 + \frac{t}{273} \right) \text{ at constant P and n}$$

Question 58

Correct gas equation is (1989)

Options:

A. $\frac{V_1 T_2}{P_1} = \frac{V_2 T_1}{P_2}$

B. $\frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$

C. $\frac{P_1 T_1}{V_1} = \frac{P_2 V_2}{T_2}$

D. $\frac{V_1 V_2}{T_1 T_2} = P_1 P_2$

Answer: B

Solution:

The ideal gas law is derived from the combination of Charles's law and Boyle's law and Avogadro's law. Charles's law states that "At constant pressure, the volume is directly proportional to temperature". It can be written as, $V \propto T$ (const. P)

Boyle's law states that "PV is constant at constant temperature". It can be written as, $PV = \text{constant}$

Avogadro's law states that "An equal volume of all gases under the same conditions of temperature and pressure contains an equal number of molecules". It can be written as, $N_1 = N_2$

By combining the above three laws, the ideal gas equation can be written as, $PV = nRT$

The ideal gas equation helps to conclude the fact that the volume of a gas depends on temperature, pressure, and the number of moles present in it. Let us have two gases of different volume, pressure, and temperature. The ideal gas equation for two different gases can be written as.

$$P_1 V_1 = nRT_1 \dots\dots (1)$$

$$P_2 V_2 = nRT_2 \dots\dots (2)$$

Dividing equation (1) and (2), we get, $\Rightarrow \frac{P_1 V_1}{P_2 V_2} = \frac{nRT_1}{nRT_2}$

Since R is gas constant and let us assume that the number of moles is the same for both gases. $\Rightarrow \frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$

Thus, the correct gas equation is $\frac{P_1 V_1}{P_2 V_2} = \frac{T_1}{T_2}$